Tianhao Li

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Education

North China University of Technology, Beijing, China **September 2020 – July 2024 (Expected)**

Undergraduate student in the branch of Information Security *GPA: 3.88/4 Score: 90.42/100 transcript*

Core Professional Courses: Applied Cryptography, Data Security, Image Processing, Principles of Computer Composition, Operating System, Reverse Engineering, Digital Forensic, Software Security, Network Attack and Defense

Skills

Programming Language: C/ModernC++(3 years+, expert), Python(3 years+, expert), Rust(1 year+, experienced), Java(1 year+, intermediate), Assembly(1 year+, experienced), Matlab(1 year+, intermediate), Shell(3 years+, intermediate)
OS & Tools: GNU/Linux(CLI&GUI, 4 years+, expert), Microsoft Windows(14 years+, expert), gcc, g++, gdb, make, cmake, vim, tmux, ssh, jupyter, etc.
Software Framework: PyTorch, PaddlePaddle, Tensorflow, TensorRT, Django, Vue.js, OpenSSL, GmSSL, etc.
Natural Language: Native Mandarin, Fluent Professional English (IELTS-6.5)
Teamwork & Documentation: Git, SVN, LaTeX, Markdown, Github, Gitlab, TortoiseSVN, Overleaf, Zotero
Miscellaneous: Driving (Since Sep 2020), Photography, Adobe PS/PR/AE/AI/AU, Cardiopulmonary Resuscitation (CPR)

Experience

China Telecommunications Group Co.,Ltd **July 2023 – September 2023**

Data Analysis Intern *Suzhou, China*

- Duty: **a)** Utilizing Python to clean, organize, and preprocess large volme of complex data in order to facilitate subsequent analytical tasks. **b)** Collaborated with colleagues to investigate and understand user's requirements, thereby adjusting data analysis methods and objectives to ensure that the analytical outcomes meet users' practical needs.
- Involved Techs & Skills: Python3, Data Cleaning, Data Analysis, Efficient Communication, Agile Process

High-Tech Innovation Institute **October 2022 – June 2023**

Research Intern *Beijing, China*

- Duty: **a)** Research and development on lightweight metahuman generation scheme based on 3D reconstruction algorithm and Alicevision framework. **b)** Research on classification of human emotion based on electroencephalogram (EEG) and Swin-Transformer. **c)** Research on early warning mechanism for aviation safety based on QAR stream data.
- Involved Techs & Skills: ML/DL, Data Mining, Python, PyTorch, Academic Writing, LaTeX, Scientific Drawing
- Mentors: Prof. Dr. Lijun Wang (RAE Fellow), Prof. Dr. Zhengping Li, Dr. Ying Li

Chenchuang Technology Development Co.,Ltd **January 2022 – March 2022**

C++ Software Development Intern *Chengdu, China*

- Duty: Develop a WebAssembly-based real-time streaming video decoding and object detection system
- Involved Techs & Skills: WebAssembly, Emscripten, FFMpeg, OpenCV, HTML/CSS/Javascript, C++

Projects

Authentication And Physical Attack Warning System Based on Gait Recognition **May 2023 – August 2023**

Team Leader, Backend Developer Demo Video 1 | Demo Video 2 | About Page

- **As team leader**: **a)** Determine team division of labor based on team members' abilities. **b)** Design system architecture. **c)** Completion of technical documentation.
- **As backend developer**: **a)** Train and deploy deep learning models in server end. **b)** Implement network module, including protocol design and server-end network programming. **c)** Implement crypto module, including secure key distribution on public network channel and encryption/decryption using asymmetric crypto system and sm2/sm3/sm4 algorithms.
- **Involved Techs & Skills**: C++11, Python3, PyTorch, PaddlePaddle, ByteTrack, PaddleSeg, GaitBase, MySQL, Digital Envelop, Digital Signature, Asymmetric Crypto System, LaTeX
- **Status**: This project has been piloted on a floor of China Telecom's office building, and has been well received by leaders and colleagues, and is still in the process of being iterated.

A Lightweight Metahuman Generation System Based on 3D Reconstruction **Jan 2023 – Apr 2023**

Backend Developer Slide | About Page

- **As backend developer**: **a)** Read the Alicevision framework documentation and its source code, and use the APIs it provides to customize the 3D reconstruction job pipeline and complete the development of the program's back-end digital human generation functions. **b)** Design software architecture, develop software interaction logic, determine server-side and client-side network protocols and implement server-side network programming. **c)** Produce simple datasets, test algorithm performance in different environments and evaluate the usability of the results in multiple dimensions, and propose an empirical method for dataset quality assessment.

- **Involved Techs & Skills**: C++11, Python3, Network Programming, 3D Reconstruction, Alicevision Framework
- **Status**: This project has been put into production use and transferred to other colleagues in the company for subsequent maintenance and improvement.

RAM File System Based on Microsoft FAT32 Format **Nov 2022 – Jan 2023**

Full-stack Developer Git | About Page

- **As full-stack developer**: **a)** Read the specification for Microsoft's FAT32 file system for top-level software engineering, including layered structure, layered functionality. **b)** Based on modern C++, STL libraries and BoostC++ libraries to implement the algorithms and functions involved in abstracting the memory hard disk layer, block read/write layer, block management layer, file management layer, and application layer. **c)** Project executable build based on CMake, data visualization based on fmt and tabulate, unit testing based on Boost. **d)** Write simple software documentation
- **Involved Techs & Skills**: C++11, Boost C++, C++ STL, CMake, Operation System, File System

Publications

- SFSWTS: EEG Emotion Recognition Based on Spatial-frequency Shifted Windows Time Self-attention Neural Network (Currently under review by Knowledge-Based System)
- **Tianhao Li**, Yujia Zheng, Haoan Zhang, Weizhi Ma and Ying Li*. "Research on Real-time Early Warning Mechanism of Aviation Safety Based on Finite State Machine Underlying in QAR Stream Data." Proceedings of the 2023 7th International Conference on Big Data and Internet of Things. 2023. (Accepted by BDIoT'23)
- Kaixuan Men,**Tianhao Li**,et al.(2022).Design and Implementation of X3DH Key Negotiation Protocol Demonstration System. Digital Technology and Application(02),228-231.

Talks

- Dive Into QAR Stream Data - A Real-time Early Warning Mechanism @ 2023 7th International Conference on Big Data and Internet of Things (BDIoT'23), Beijing, China. [slide]
- Hyper-Realistic MetaHuman - The Bridge to the Metaverse in Web 3.0 @ High-tech Innovation Institute, Beijing, China. [slide]
- Tutorial of C Programming Language @ School of Information, North China University of Techonology, Beijing, China. [slide][src][video]
- An Overview of Image Steganography @ Department of Information Security, North China University of Techonology, Beijing, China. [slide]
- A Simple Introduction to Spectrum Map @ Department of Information Security, North China University of Techonology, Beijing, China [slide]

Honors and Awards

National Contests

- (Aug.2023, National) Third Prize in 16th CISCN Security Project Contest [report][slide][video]
- (Apr.2023, National) First Prize in 13th MathorCup Mathematical Modeling Challenge [certificate][paper]
- (Jun.2021, National) Third prize in 12th Lanqiao Software Development and Algorithm Contest [certificate]

Non-National Contests

- (Jun.2023, Regional) Third Prize in 16th CISCN AWDP (Attack with Defense Plus) Contest
- (Apr.2023, Provincial) First Prize in 14th Lanqiao Software Development and Algorithm Contest [certificate]
- (Jun.2022, Regional) Third Prize in 15th CISCN CTF (Capture the Flag) Contest [certificate]
- (Apr.2021, Provincial) First Prize in 12th Lanqiao Software Development and Algorithm Contest [certificate]

Scholarships

- (2022-2023) Received scholarships for outstanding students, academic, innovation, athletes, moral, ¥8800 in total.
- (2021-2022) Received scholarships for outstanding students, academic, innovation, athletes, moral, ¥6600 in total.
- (2020-2021) Received scholarships for academic, innovation, ¥1800 in total.

Certifications

- Certificate of completion of Artificial Intelligence course from *China Computer Federation (CCF)* [certificate]
- Certificate of oral presentation of 2023 7th International Conference on Big Data and Internet of Things (BDIoT'23) from *BDIoT'23 conference committees* [certificate]
- Position Competency Assessment Certificate of Industry and Information Technology Talents from *Ministry of Industry and Information Technology of the People's Republic of China* [certificate]

Open Source Activities

- I am contributor of OpenGait(470+stars, 110+forks) and All-in-One-Gait(20+stars, 5+forks).
- I am owner of OpenNCUT, VigenereCracker, simple_ram_fs, Histogram-Equalization-Specification-Demo, DES_cpp_demo.

Extracurricular Activities

- Mountain Biking: See POV videos on bilibili channel. Top 9 racer in 2023 GDL Thaiwoo downhill race.
- Road/Trail Running and Hiking: Finisher in 2023 Chongli 168 Ultra Trail(Biggest in Asia) DTC-100KM race.
- Photography and Off-road Driving: Keen on wildlife photography and exploring the unknown.
- Voluntary service: nearly 400 hours were recorded by Beijing Volunteer Association (As of July 2023)

李天昊

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教育背景

北方工业大学, 中国, 北京 **2020年9月 – 2024年7月（预计）**

本科在读 - 信息安全 绩点: *3.88/4* 成绩: *90.42/100* 成绩单

核心专业课程: 应用密码学, 数据安全, 数字图像处理, 计算机组成原理, 操作系统, 逆向工程, 数字取证, 软件安全, 网络攻防

专业技能

编程语言: C/现代C++(3年, 熟练), Python(5年, 熟练), Rust(1年, 使用过), Java(1年, 中等), Assembly(1年, 使用过), Matlab(1年, 中等), Shell(3 years+, 中等)
操作系统与工具: GNU/Linux(命令行与图形界面, 4年, 熟练), Microsoft Windows(14年, 熟练), gcc, g++, gdb, make, cmake, vim, tmux, ssh, jupyter, etc.
软件框架: PyTorch, PaddlePaddle, Tensorflow, TensorRT, Django, Vue.js, OpenSSL, GmSSL, etc.
自然语言: 普通话, 流利的专业英语 (IELTS-6.5)
团队合作与文档撰写: Git, SVN, LaTeX, Markdown, Github, Gitlab, TortoiseSVN, Overleaf, Zotero
杂项: 驾驶(从2020年9月), 摄影, Adobe PS/PR/AE/AI/AU, 心肺复苏(CPR)

工作经历

中国电信股份有限公司 **2023年7月 – 2023年9月**

数据分析实习 中国，苏州

- 工作职责: **a)** 利用 Python 对大量复杂数据进行清理、组织和预处理，以便于后续的分析任务。 **b)** 与同事合作，调查和了解用户的要求，从而调整数据分析方法和目标，确保分析结果符合用户的实际需要。
- 技术与技能: Python3, 数据清洗, 数据分析, 高效沟通, 敏捷过程

高精尖创新研究院 **2022年10月 – 2023年6月**

科研实习 中国，北京

- 工作职责: **a)** 研发基于三维重建算法和Alicevision框架的轻量级数字人生成方案。 **b)** 基于脑电图和Swin Transformer的人类情感分类研究。 **c)** 基于QAR流数据的航空安全预警机制研究。
- 技术与技能: 机器学习与深度学习方法, 数据挖掘技术, Python3, PyTorch, 学术写作, LaTeX, 科研绘图
- 导师: 王立军院士, 李争平教授, 李颖博士

辰创科技发展有限公司 **2022年1月 – 2022年3月**

C++软件开发实习 中国, 成都

- 工作职责: 开发基于WebAssembly的实时流媒体视频解码和目标检测系统
- 技术与技能: WebAssembly, Emscripten, FFMpeg, OpenCV, HTML/CSS/Javascript, C++

项目实践

基于步态识别的身份验证和近源渗透预警系统 **2023年5月 – 2023年8月**

团队领导者, 后端开发者 演示视频1 | 演示视频2 | 介绍页

- **作为团队领导者**: **a)** 根据团队成员的能力确定团队分工。 **b)** 设计系统技术架构。**c)** 完成技术文档的撰写。
- **作为后端开发者**: **a)** 在服务器端训练和部署深度学习模型。 **b)** 实现网络模块，包括协议设计和服务器端网络编程。 **c)** 实现加密模块，包括在公共网络通道上安全分发密钥，以及使用非对称加密体制和sm2/sm3/sm4算法进行加密与解密。
- **技术与技能**: C++11, Python3, PyTorch, PaddlePaddle, ByteTrack, PaddleSeg, GaitBase, MySQL, 数字信封技术, 数字签名技术, 非对称密码体制, LaTeX
- **项目状态**: 本项目曾在中国电信办公楼某层进行试运行，获得了领导和同事的好评，仍在不断迭代中。

基于三维重建技术与Alicevision框架的轻量级数字人生成系统 **2023年1月 – 2023年4月**

后端开发者 汇报幻灯片 | 介绍页

- **作为后端开发者**: **a)** 阅读Alicevision框架文档和源码，利用其提供的编程接口，客制化三维重建作业流水线，完成程序后端数字人生成功能开发。 **b)** 设计软件架构，制定交互逻辑，确定服务端与客户端的网络协议并实现服务端网络程序设计。 **c)** 制作简单数据集，在不同环境下测试算法性能并从多个维度评估结果可用度，并基于经验提出一种数据集质量评估方法。
- **技术与技能**: C++11, Python3, 网络程序设计，三维重建，Alicevision框架
- **项目状态**: 本项目已经投入生产使用并移交给公司其他同事进行后续维护与完善。

基于微软FAT32文件系统格式的内存文件系统的设计与实现 **2022年11月 – 2023年1月**

全栈开发者 Git | 介绍页

- **作为全栈开发者**: **a)** 阅读微软FAT32文件系统的技术手册，进行软件工程顶层设计，包括分层结构、分层功能。 **b)** 基于现代C++、STL库以及BoostC++库实现抽象内存硬盘层、块读写层、块管理层、文件管理层、应用层涉及到的各算法与函数。 **c)** 基于CMake进行项目可执行文件构建，基于fmt与tabulate进行数据可视化，基于Boost.Test进行单元测试。 **d)** 撰写简单的软件文档。
- **技术与技能**: C++11, Boost C++, C++ STL, CMake, 操作系统, 文件系统

文章发表

- SFSWTS: EEG Emotion Recognition Based on Spatial-frequency Shifted Windows Time Self-attention Neural Network (Currently under review by Knowledge-Based System)
- **Tianhao Li**, Yujia Zheng, Haoan Zhang, Weizhi Ma and Ying Li*. "Research on Real-time Early Warning Mechanism of Aviation Safety Based on Finite State Machine Underlying in QAR Stream Data." Proceedings of the 2023 7th International Conference on Big Data and Internet of Things. 2023. (Accepted by BDIoT'23)
- Kaixuan Men,**Tianhao Li**,et al.(2022).Design and Implementation of X3DH Key Negotiation Protocol Demonstration System. Digital Technology and Application(02),228-231.

公开演讲

- Dive Into QAR Stream Data - A Real-time Early Warning Mechanism (English) @ 2023第七届大数据与物联网国际会议 (BDIoT'23), 中国, 北京 [幻灯片]
- Hyper-Realistic MetaHuman - The Bridge to the Metaverse in Web 3.0 (English) @ 高精尖创新研究院, 中国, 北京 [幻灯片]
- C语言期末教程 @ 北方工业大学信息学院, 中国, 北京 [幻灯片][源码][视频]
- An Overview of Image Steganography (English) @ 北方工业大学信息安全系, 中国, 北京 [幻灯片]
- A Simple Introduction to Spectrum Map (English) @ 北方工业大学信息安全系, 中国, 北京 [幻灯片]

荣誉奖项

国家级竞赛

- (2023年8月, 国家级) 第十六届全国大学生信息安全竞赛作品赛三等奖 [报告][幻灯片][演示视频]
- (2023年4月, 国家级) 第十三届MathorCup数学建模挑战赛一等奖 [证书][论文]
- (2021年6月, 国家级) 第十二届蓝桥杯全国软件和信息技术专业人才大赛三等奖 [证书]

非国家级竞赛

- (2023年6月, 区域级) 第十六届全国大学生信息安全竞赛实践赛(AWDP)三等奖
- (2023年4月, 省部级) 第十四届蓝桥杯全国软件和信息技术专业人才大赛一等奖 [证书]
- (2022年6月, 区域级) 第十五届全国大学生信息安全竞赛实践赛(CTF)三等奖 [证书]
- (2021年4月, 省部级) 第十二届蓝桥杯全国软件和信息技术专业人才大赛一等奖 [证书]

奖学金

- (2022-2023学年) 荣获优秀学生、学习、创新实践、体育、思想品德奖学金, 共计8800元。
- (2021-2022学年) 荣获优秀学生、学习、创新实践、体育、思想品德奖学金, 共计6600元。
- (2020-2021学年) 荣获学习、创新实践奖学金, 共计1800元。

荣誉证书

- CCF群星计划之人工智能科普培训结业证书 来自 中国计算机学会(CCF) [证书]
- 第七届大数据与物联网国际会议报告证书 来自 BDIoT'23大会委员会 [证书]
- 工业和信息化人才岗位能力评价证书 来自 中华人民共和国工业和信息化部 [证书]

开源活动

- 我是OpenGait(470+stars, 110+forks)和All-in-One-Gait(20+stars, 5+forks)的开源贡献者。
- 我是OpenNCUT, VigenereCracker, simple_ram_fs, Histogram-Equalization-Specification-Demo, DES_cpp_demo等开源仓库的拥有者。

课外活动

- 山地自行车: 可以访问B站主页观看视频。2023GDL太舞速降赛排名前九。
- 跑步与徒步: 荣获2023崇礼168超级越野赛DTC百公里组完赛奖。
- 摄影与驾驶: 热衷于野生动物摄影和探索未知世界。
- 志愿服务: 志愿北京平台记录近400小时。